

# The High-Redshift Accretion Atlas

Reproducible Catalogue and Complete Visual Growth Atlas, v3

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27 August 2026

## Abstract

We present the v3 provenance-aware atlas of reported accreting black holes at  $z \geq 4$ . The catalogue contains 234 literature measurements representing 219 physical objects in 218 host systems. Its science layer uses 209 measurements and 196 preferred physical-object rows with canonical numeric black-hole masses; 23 catalogue objects without such masses remain explicit exclusions. The atlas keeps heterogeneous evidence classes and mass methods separate, retains alternate measurements, and treats global ranks as navigation rather than demographic inference. A full-source audit corrects the Scholtz et al. JADES narrow/high-ionization subset to 21 objects at  $z \geq 4$  and documents the paper/table count discrepancy while retaining all source history. Under the reference assumptions  $z_{\text{seed}} = 30$ ,  $\epsilon = 0.1$ , and no merger boost, luminous quasars occupy the highest growth-pressure region. These diagnostics measure pressure under stated assumptions; they do not identify a unique seed or accretion history.

## 1 Purpose and interpretation boundary

The atlas asks which reported objects place the strongest pressure on simple black-hole growth histories, and which measurements most affect that conclusion. It does not estimate a population mass function from the pooled catalogue. The input literature spans broad-line AGN selected by several JWST surveys [2, 3, 4, 5, 6, 7], luminous-quasar comparison samples [10, 12], host-selected broad-line candidates [9], an X-ray evidence history [13, 15], and JADES narrow/high-ionization candidates [16]. Selection functions, evidence strength, and mass estimators are therefore retained as analysis strata.

## 2 Catalogue construction and provenance

Every measurement has a stable measurement ID, physical-object ID, and host system ID. Source-native values, adopted canonical values, uncertainty kinds, method tags, URLs, available archive checksums, and identity decisions are preserved. Missing historical provenance is explicit: the 23 frozen-v1 Juodžbalis/JADES rows predate archive hashing and retain a documented unavailable checksum and extraction-date limitation rather than a fabricated value. Auxiliary coordinate and context inputs are attributed separately to the E-XQR-30 survey catalogue [11], the THRILS programme catalogue [8], the UHZ1 spectroscopy paper [14], and the versioned JADES DR3 GOODS-S prism catalogue. Alternate measurements are never silently averaged or deleted. Exactly one preferred measurement controls each object-level status in v3, while the complete history of measurement-level evidence statuses and bases remains in the object table. This resolves the case in which a weaker alternate candidate row had conservatively downgraded an otherwise secure preferred measurement.

The Scholtz et al. source audit deserves specific note. The paper reports 42 objects, while its source-native sample table contains 41 rows. Parsing that full table gives exactly 21 rows at  $z \geq 4$ . The pre-correction extraction contained 20 of them; canonical v3 includes the omitted JADES-NS-GS00099671

row ( $z = 5.936$ ) with its published host and bolometric observables, but no invented black-hole mass. The 21 candidates include one identity-linked broad-line object, so the current physical-object catalogue contains 20 narrow-line candidates from this source.

**Table 1:** v3 scale and explicit analysis subsets.

| Product or subset                                           | Rows / objects  |
|-------------------------------------------------------------|-----------------|
| Measurements / physical objects / host systems              | 234 / 219 / 218 |
| Source-local observables                                    | 1,106           |
| Growth-eligible measurements / objects                      | 209 / 196       |
| Primary measurements / objects                              | 182 / 171       |
| Broad-line / luminous-quasar eligible objects               | 112 / 84        |
| Catalogue-only physical objects                             | 23              |
| Alternate-measurement sensitivity pairs                     | 13              |
| Follow-up rows / ranked rows                                | 219 / 196       |
| Source-family caveat rows                                   | 11              |
| Per-object $f_{\text{Edd}}$ -mass/seed-redshift-mass panels | 438             |

A supplementary [full-assumption growth-track companion](#) preserves the historical v1 visual grid: three seed masses crossed with  $f_{\text{Edd}} = 0.3, 1, 2$ , four constant radiative efficiencies, and two merger boosts, for 72 reference curves in total. This denser view is an assumption comparison, not a fit or a demographic model.

An additional [uncertainty-filtered companion](#) retains the same 72 curves while excluding only the seven luminous quasars whose maximum reported black-hole-mass uncertainty exceeds 0.5 dex. The exact selection is preserved in [the filter audit table](#); all other numerical objects and all 23 no-inference markers remain visible. This is a display-specific sensitivity view and does not alter the canonical catalogue or rankings.

### 3 Growth model and uncertainty

The reference model follows exponential radiatively efficient growth [1],

$$M_{\text{BH}}(t_{\text{obs}}) = M_{\text{seed}} \exp \left[ f_{\text{Edd}} \left( \frac{1 - \epsilon}{\epsilon} \right) \frac{t(z_{\text{obs}}) - t(z_{\text{seed}})}{t_{\text{Edd}}} \right] B_{\text{merge}}, \quad (1)$$

with  $t_{\text{Edd}} \simeq 0.45$  Gyr. The main tables invert this relation to report required lifetime-average  $f_{\text{Edd}}$  for fixed seed masses and required  $M_{\text{seed}}$  for fixed  $f_{\text{Edd}}$ . The baseline uses  $z_{\text{seed}} = 30$ ,  $\epsilon = 0.1$ , and  $B_{\text{merge}} = 1$ .

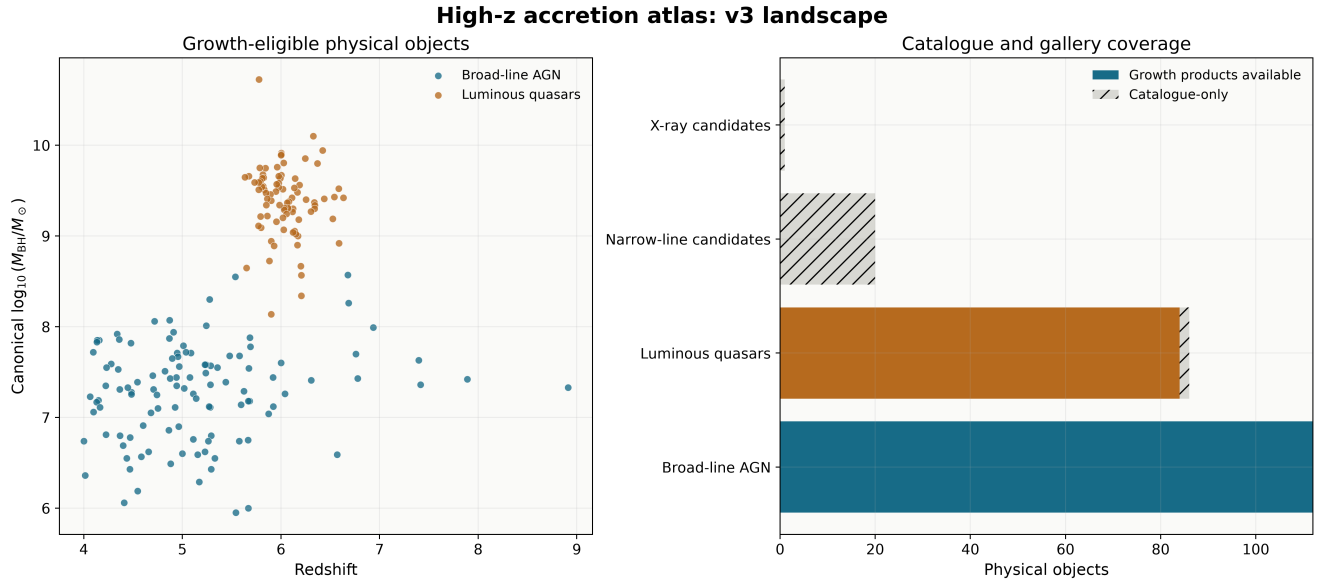
Reported asymmetric statistical uncertainties in  $\log_{10} M_{\text{BH}}$  are propagated with 10,000 deterministic, per-object split-side draws. Method-scale systematic envelopes remain separate from the statistical sampling. A reported current Eddington ratio is not substituted for the lifetime-average growth rate.

### 4 Current results

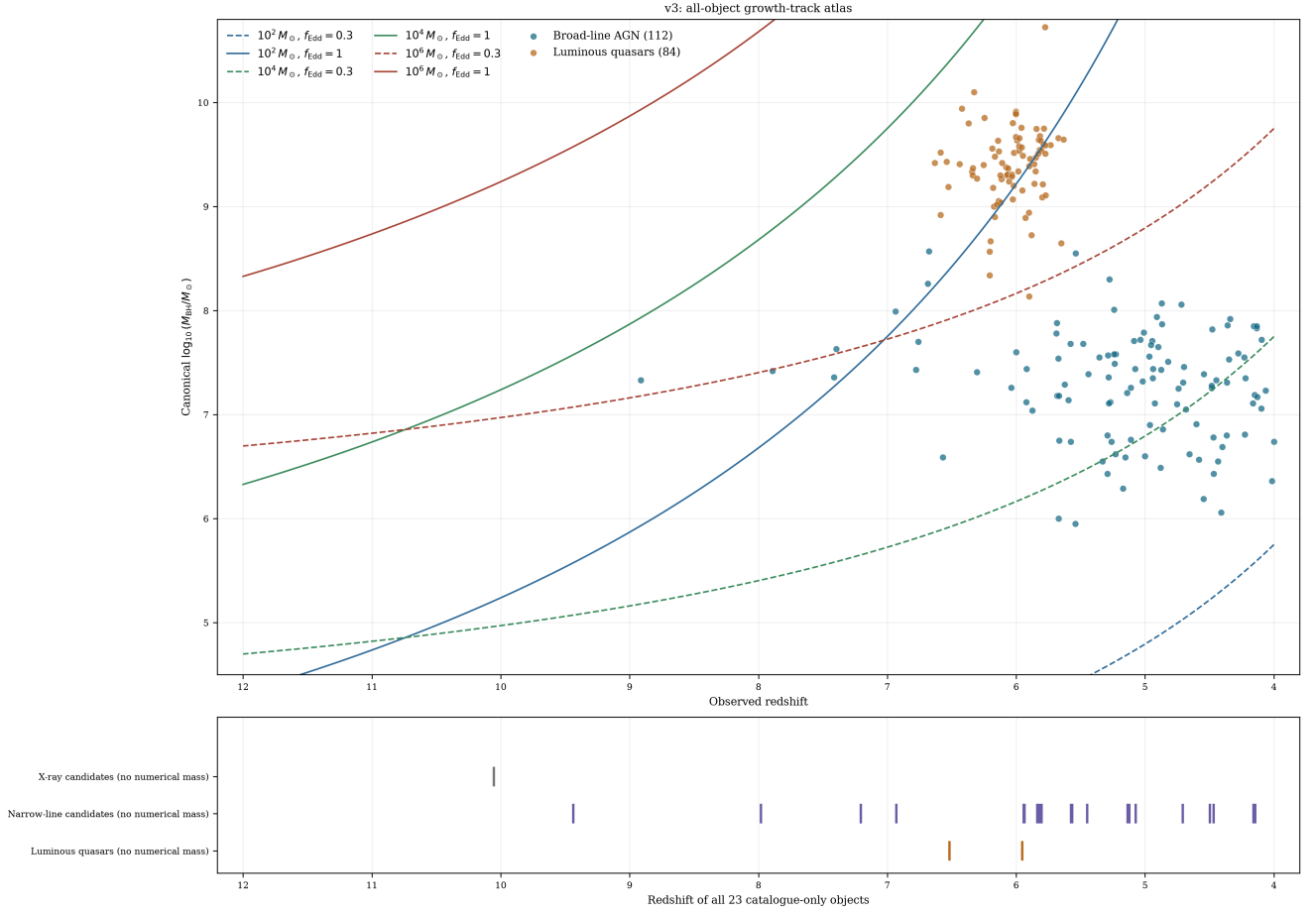
The global navigation view places luminous quasars at the highest reference growth pressure. J1148+5251 is first in the point and uncertainty-aware views, with median required  $f_{\text{Edd}} = 1.214$  for a  $10^2 M_{\odot}$  seed. SDSS J0100+2802 and PSO J231-20 follow. The top eight uncertainty-ranked objects have unit Monte Carlo probability, to the stored precision, of requiring  $f_{\text{Edd}} > 1$  under the same light-seed reference. These

results express tension with that reference history, not a claim of literally continuous super-Eddington accretion.

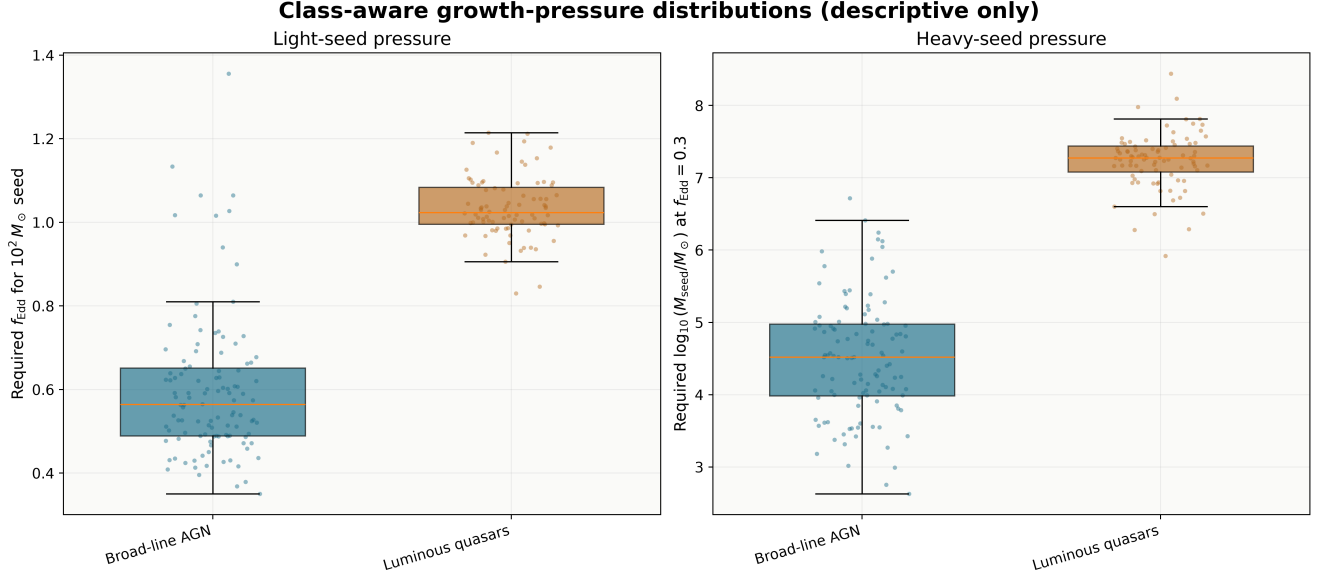
Within-class and within-class–mass-method ranks are the supported comparison scopes. All outputs explicitly forbid pooled demographic inference without a selection-function and completeness model. The alternate-measurement table contains 13 comparisons; it shows how retained measurement choice changes  $\log_{10} M_{\text{BH}}$  and required  $f_{\text{Edd}}$  while leaving the canonical measurement choice auditable. The follow-up matrix retains all 219 objects, ranks the 196 growth-eligible objects, and leaves the other 23 visibly unranked. A separate 11-row caveat table gives one compact provenance and interpretation summary for every admitted source family.



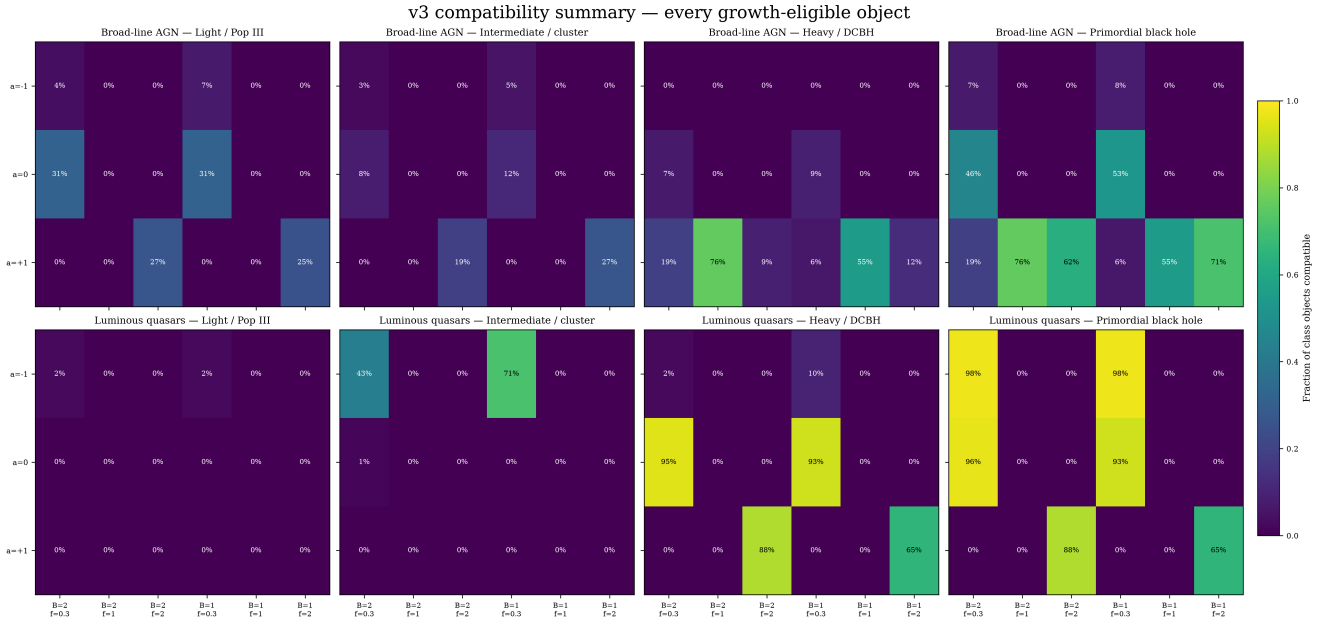
**Figure 1:** Current catalogue landscape. Left: all 196 growth-eligible physical objects in black-hole-mass–redshift space. Right: growth-product coverage for all 219 objects. The 23 catalogue-only objects remain available for provenance and evidence analyses but are not assigned numeric growth products.



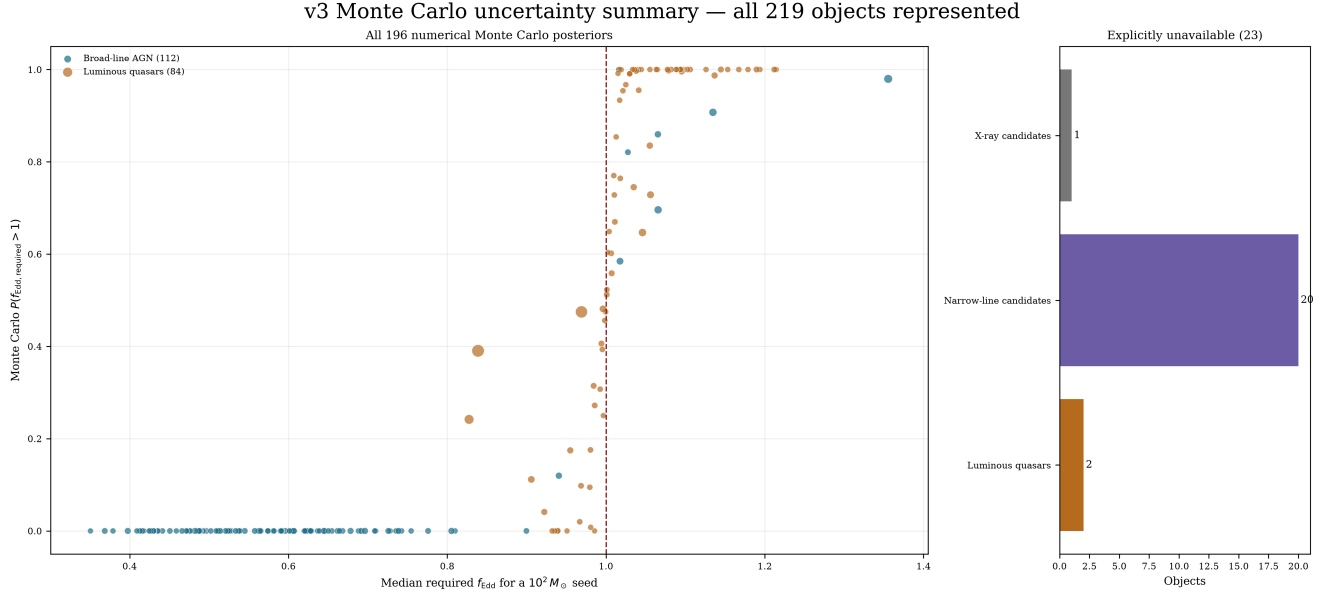
**Figure 2:** Complete v3 growth-track view. All 196 objects with supported canonical black-hole masses are plotted against reference tracks. The lower panel retains the redshift locations of all 23 catalogue-only objects for which a numerical growth history cannot be inferred. Thus every physical object is visible without assigning unsupported masses.



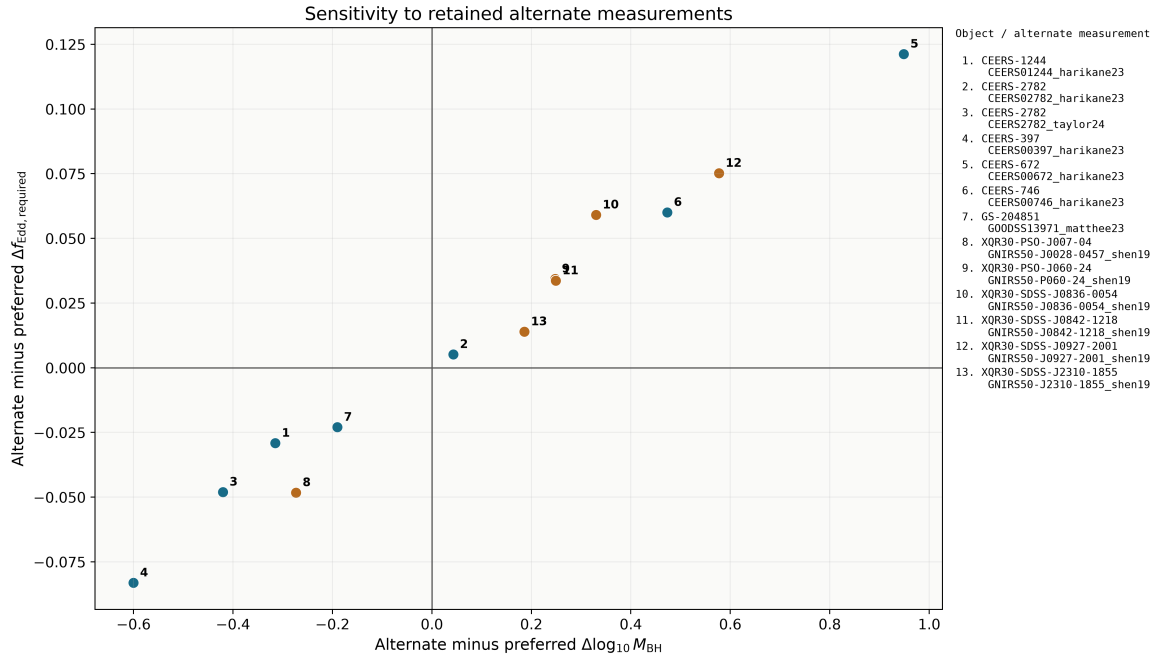
**Figure 3:** Descriptive class-aware growth-pressure distributions. The luminous quasar comparison sample and faint broad-line AGN are shown separately because their selection functions and mass regimes differ. Boxes and points summarize the canonical object-level table; they are not completeness-corrected demographic estimates.



**Figure 4:** Compatibility summary using every one of the 196 growth-computable objects. Panels retain object class and seed family while spanning spin, merger boost, and average accretion rate. Percentages are descriptive within-class fractions, not pooled demographic estimates. The complete object-by-object matrix, including all 23 unavailable objects, is included in the supplement.



**Figure 5:** Complete Monte Carlo uncertainty summary. Every one of the 196 numerical posteriors is plotted; point size encodes the 16th–84th percentile width. The adjacent accounting panel retains all 23 catalogue-only objects. The full supplement atlas gives every object label and interval.



**Figure 6:** Sensitivity to 13 retained alternate measurements. Numbered points are keyed to stable object and measurement IDs, avoiding ambiguous overlapping labels. The plot quantifies measurement-choice sensitivity rather than treating alternates as independent objects.

## 5 Reproducibility and canonical artifacts

The Python 3.12 direct environment dependencies are pinned. Canonical dataset manifests verify exact catalogue, science, and figure membership and SHA-256 hashes. In-memory reproduction compares generated tables against checked-in CSVs without rewriting them. Regression tests enforce cardinalities, source anchors, identity links, evidence aggregation, uncertainty boundaries, ranking policy, figure resolution, and every canonical gallery path. Only v1/v2/v3 products and manifests are public; the source-admission bridge required for reconstruction is isolated as internal compatibility code. Superseded release-era artifacts remain recoverable from Git history.

The final figure layer provides paper-facing summaries and a complete current gallery. Every one of the 219 objects has an  $f_{\text{Edd}}$ -mass panel and a seed-redshift-mass panel under the final gallery path. For the 23 unavailable objects these are explicit no-inference panels. Complete  $f_{\text{Edd}}$ -mass, compatibility, and Monte Carlo atlas figures preserve every object label for close zooming. Growth tracks are retained as combined all-object figures rather than individual gallery panels; the supplementary full-assumption view preserves all 72 historical v1 reference curves for direct comparison with v3, while its uncertainty-filtered companion makes the dense comparison readable without the seven luminous-quasar error-bar outliers.

## 6 Limitations and next scientific work

- The catalogue is heterogeneous and cannot support pooled demographics without explicit selection and completeness models.
- Virial masses are method-dependent; available method systematics are reported separately and are not full posterior reconstructions.
- Narrow-line and X-ray candidates without canonical numeric masses are valuable evidence records but cannot enter the current growth ranking.
- The reference model compresses time-variable accretion, feedback, mergers, and radiative-state changes into effective parameters.
- Identity review uses deterministic candidates and documented decisions; crowded or imprecise cases may ultimately require probabilistic crossmatching.

The next scientific milestone is no longer basic catalogue assembly. It is a selection-aware inference layer: explicit survey selection functions, completeness modeling, method-calibrated hierarchical uncertainty, and a value-of-information refinement of the existing follow-up matrix tied to which measurement would most change each object’s physical interpretation.

## 7 Conclusion

v3 is a complete, internally consistent research dataset: the current catalogue, class-aware science tables, paper figures, full gallery coverage, provenance correction, evidence policy, and automated verification all refer to the same dataset version. It provides reproducible observational triage across 219 objects while keeping unsupported demographic and formation-channel claims outside the scope of the data product.

## References

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